Curated Research Articles

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- **High-Entropy Design for Improving Ionic Conductivity and Air Stability of Lithium Argyrodites for All-Solid-State Lithium-Sulfur Batteries** — score: 1.000 This study introduces a novel high-entropy design strategy for lithium argyrodites, enhancing both ionic conductivity and air stability, which are crucial for their use in all-solid-state lithium-sulfur batteries. By substituting elements in the lithium argyrodite structure, the researchers achieved a significant improvement in ionic conductivity (up to 11.4 mS cm−1) and demonstrated better chemical stability against moisture, paving the way for more reliable and efficient battery applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Ni-Rich Cathodes for High-Energy All-Solid-State Batteries Suppressing Phase-Gradients** — score: 1.000 This study explores the impact of microstructural tuning through multivalent doping on the electrochemical performance of Ni-rich cathodes in sulfide-based all-solid-state batteries (ASSBs). By enhancing the microstructure to feature radially aligned particles, particularly with W doping, the researchers demonstrate improved Li-ion diffusion, reduced microcracking, and diminished phase gradients, leading to increased capacity and cycling stability. These findings contribute valuable design principles for optimizing Ni-rich cathodes in high-energy ASSBs. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Separating Hydrogen Evolution From Electroplating in Zn Metal Batteries Using Scanning Electrochemical Microscopy** — score: 1.000 The article discusses the use of scanning electrochemical microscopy (SECM) to differentiate hydrogen evolution from other degradation reactions during the electroplating of zinc in zinc metal batteries. The study finds that while hydrogen evolution occurs at similar potentials to zinc plating, its impact on irreversible charge loss is minor, highlighting the need to understand interfacial mechanisms to enhance battery performance and lifespan. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Timescale-Decoupled Polarization Kinetics Guide the Design of Polarization-Resistant Cathode Hosts for Zinc–Iodine Batteries** — score: 1.000 The article presents a new framework to differentiate between various types of voltage polarization in zinc–iodine batteries, identifying concentration polarization as the main challenge at high iodine loadings. A novel polarization-resistant cathode, Fe@PG, is designed with a porous graphene structure that optimizes mass transport and electron flow, resulting in significantly reduced polarization losses and impressive performance metrics, including high capacity retention after 3000 cycles. This work lays the groundwork for improving polarization management in similar battery technologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Understanding the Electrolyte-Hard Carbon Interphase Synergy in Sodium-Ion Batteries: From Mechanistic Insights to Design Strategies** — score: 1.000 The article explores the synergistic effects of electrolyte and hard carbon engineering on the formation and stability of the solid electrolyte interphase (SEI) in sodium-ion batteries. It highlights how modifications in electrolyte composition and hard carbon structures influence SEI properties, leading to improved performance and resilience in batteries. Proposed design strategies aim to enhance the interfacial chemistry, suggesting avenues for future research to create robust and efficient sodium-ion battery systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Multi-Selective Dominant Coordination in Multi-Salt High-Entropy Electrolytes Toward High-Performance Silicon Anodes** — score: 1.000 The article discusses the innovative use of high-entropy electrolytes (HEEs), specifically through a proposed mechanism called multi-selective dominant coordination, to enhance the performance of lithium-ion batteries. By employing a variety of salt combinations, the study demonstrates that HEEs lead to improved ionic conductivity and a robust solid electrolyte interphase (SEI), resulting in a significant capacity retention rate in pouch cells after 200 cycles. This advancement addresses a critical challenge in balancing rapid ion transport and durable interfacial chemistry in battery design. Journal: *Wiley: Advanced Energy Materials: Table of*

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- **Regulating Interfacial Electron Density of Carbon for Stable and High-Performance Batteries** — score: 1.000 This review discusses the dual role of conductive carbon in batteries, where it facilitates charge transport but can also instigate detrimental parasitic reactions due to fluctuating interfacial electron density. It introduces an "electronic stability window" to understand these reactions and presents design strategies—such as Fermi-level engineering and defect-state optimization—to regulate electron density, ultimately improving battery performance and longevity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Decoupling Thermodynamic and Kinetic Moisture Instability Reveals a Critical Humidity Threshold for Sodium Thiophosphate Solid Electrolyte** — score: 1.000 The study establishes a moisture-stability framework for sodium thiophosphate (Na3PS4), highlighting a critical humidity threshold of 15% relative humidity (RH) and a heat treatment temperature of 80°C that influence its recoverability after exposure to moisture. It differentiates between thermodynamic and kinetic factors affecting moisture-induced degradation, indicating that Na3PS4 can undergo reversible hydration without significant hydrolysis at lower humidity levels, thereby offering insights for improving manufacturing processes in solid-state batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Topologically Robust New Silica Phase Decoupling Intrinsic Zero-Strain Feature and High-Capacity Li+ Storage** — score: 1.000 The article presents a novel topologically robust silica phase (TRS) that effectively resolves the conflict between high lithium-ion storage capacity and minimal structural deformation. By utilizing a rigid Si–O framework with large cavities, TRS achieves a remarkable capacity of approximately 800 mAh g−1 while maintaining only about 1.5% strain, surpassing the performance of traditional silicon-based and zero-strain anodes, thereby offering a promising approach for future lithium-ion battery anode design. Journal: *Wiley: Small Methods: Table of Contents*
- **Unraveling Phase Evolution Pathways and Sequential Multi-Metal Reduction Mechanisms in Over-Lithiated NCM Cathodes** — score: 1.000 This study presents a comprehensive model for over-lithiation in NCM cathodes, detailing two distinct reaction phases: an initial intercalation phase allowing for excess lithium through sequential metal reduction, followed by a conversion phase that leads to structural degradation. The findings elucidate the complex redox interactions and highlight the mechanical strain resulting from over-lithiation, providing valuable insights for enhancing the energy density of lithium batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **High-Spin 3d Orbital Configuration Driven by Lattice Strain-Induced Electronic Modulation in Co4N/Co2P Composites Enables Metal–Sulfur Hybridization for Efficient Polysulfide Conversion** — score: 1.000 The study presents a Co4N/Co2P composite that leverages lattice strain to induce a transition to high-spin Co 3d electron states, improving polysulfide interaction in lithium-sulfur batteries. By modulating the electronic structure through interfacial strain effects, the composite enhances charge transfer and adsorption, resulting in high electrochemical performance with significant capacity retention over cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Recent Interface Interactions Engineering in MXene Composite Cathodes for Enhanced Zinc-Ion Batteries** — score: 0.900 This review presents a comprehensive framework for understanding MXene composite cathodes in aqueous rechargeable zinc-ion batteries, focusing on the critical aspects of interfacial bonding strength and spatial geometry. By examining recent advancements and establishing a systematic approach, the work clarifies how strong chemical interactions at the interfaces contribute to improved cycling stability and performance. The authors propose an integrated roadmap for further developing durable and efficient zinc-ion battery technologies. Journal: *Wiley: Small Methods: Table of Contents*
- **Grain-Boundary-Rich SEI Synergizing with an H2O-Poor EDL Enables Multidimensional Interfacial Regulation for Stable Zn Anodes** — score: 0.900 The study presents the use of a trace

zinc hyaluronate (HA-Zn) additive to create a robust, H2O-poor electric double layer and a grain-boundary-rich solid-electrolyte interphase on zinc anodes, enhancing ion transport and stability. This innovative approach leads to impressive performance metrics, achieving over 4,750 hours of lifespan in symmetric cells and maintaining 86.34% capacity retention in pouch batteries after 3,300 cycles, addressing significant interfacial challenges in aqueous zinc-ion batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Synergistic Anode Engineering for Aqueous Aluminium Ion Batteries: From Mechanistic Understanding to Computationally Accelerated Design** — score: 0.900 The article reviews the development of aqueous aluminium-ion batteries (AAIBs) by focusing on anode engineering to address challenges at the anode-electrolyte interface, which affect battery stability. It highlights innovative strategies such as alloying, surface engineering, and electrolyte modification, exploring how these methods can simultaneously mitigate multiple failure modes while advancing the mechanistic understanding necessary for optimizing performance in sustainable energy applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Diffusionless mechanisms produce diffusion-like electrochemistry in battery particles** — score: 0.800 The article reveals that low-frequency overpotentials in Li-ion battery particles are not solely due to solid diffusion, as previously thought. Instead, it highlights how electrolyte penetration activates the interior surfaces of agglomerate particles in a transmission-line manner, resulting in diffusion-like electrochemical behavior without actual diffusion. Journal: *Nature Energy*
- **Solution-mediated reversible anion storage enables high-power organic batteries** — score: 0.800 The study reveals that p-type organic electrodes in batteries utilize a solution-mediated redox pathway, enhancing ion transport and reaction efficiency. This novel mechanism, characterized by the formation of transient solvated intermediates and subsequent recrystallization, accounts for the observed high power output and longevity of the batteries, outperforming traditional solid-state methods. Journal: *Joule*
- **Hard carbon anodes for sodium-ion batteries: A critical review from structural and mechanistic insights to practical optimization strategies** — score: 0.800 This critical review examines the properties and performance of hard carbon anodes in sodium-ion batteries, providing insights into their structural characteristics and underlying mechanisms. It also discusses practical optimization strategies to enhance their efficiency and sustainability in energy storage applications. Journal: *ScienceDirect Publication: Nano Energy*
- **Ether-Based Electrolytes and the Solvation Structure in Sodium Ion Batteries** — score: 0.800 This review article discusses the role of ether-based electrolytes in enhancing the performance of sodium-ion batteries, emphasizing their unique solvation structures and co-intercalation behaviors. It addresses the current challenges in understanding the dynamic evolution of these structures during electrochemical processes and summarizes recent advancements in characterizing solvation and its impact on interfacial chemistry and overall battery performance. The authors highlight the need for further research to improve the rational design of high-performance electrolytes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Edge-Synergistic Ni Single Atoms on MoS2 Nanoflake Islands Enable Tandem Polysulfide Electrocatalysis in Lithium–Sulfur Batteries** — score: 0.800 The article discusses the development of nickel single atom catalysts supported on MoS2 nanoflake islands, which enhance electrocatalytic activity for polysulfide conversion in lithium-sulfur batteries. This design aims to improve battery efficiency by facilitating tandem reactions at the edges of the nanostructures, ultimately contributing to better overall performance of the batteries. Journal: *ScienceDirect Publication: Nano Energy*
- **Fluoride-Driven Interfacial Engineering: Architecting Functional Layers for High-Performance Dendrite-Free Li Metal Batteries** — score: 0.800 The article reviews the use of metal fluorides (MxF) in lithium metal batteries (LMBs) to combat dendritic growth and enhance battery performance through the creation of robust solid electrolyte interphases (SEIs). It explores their role in various battery components, including artificial interphase layers and electrolyte additives, and

emphasizes their potential for improving cycle life and energy density, as well as their practical application in pouch cells. Future commercialization prospects for these materials in the battery industry are also discussed. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Fluorine-Aligned Electrostatic Gatekeepers Regulating Polysulfide Migration for Stable Lithium–Sulfur Batteries (Adv. Energy Mater. 31/2026)** — score: 0.800 The article discusses a novel electrostatic separator for lithium-sulfur batteries, utilizing a -PVdF-anchored MXene scaffold. This design exploits aligned C–F dipoles to generate an electrostatic field that repels lithium polysulfide anions, effectively preventing their migration and enhancing battery stability by keeping redox intermediates confined within the cathode. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Cycling Performance of Sodium-Metal Chloride Battery Cells and Modules Based on Iron and Zinc** — score: 0.800 The study investigates high-temperature sodium-metal chloride cells using an Fe and Zn cathode as a cost-effective and environmentally friendly alternative to Ni-based cells. The modified chemistry demonstrates improved cycling performance and stability at 300°C, achieving near-theoretical capacity and high efficiency while addressing aging issues through voltage limitation in multi-cell modules. Post-cycling analysis reveals material interactions and suggests pathways for enhancing durability and mitigating capacity decline. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Phase-Engineered 1T/2H-WS2 Heterostructures in Hybrid Carbon Networks for High-Rate and Durable Sodium-Ion Storage** — score: 0.800 The study demonstrates a successful phase-engineering approach to create a stable 1T/2H-WS2 heterostructure integrated within a hybrid rGO/MWCNT carbon framework, enhancing sodium-ion storage performance. This innovative composite exhibits remarkable rate capability, impressive long-term cycling stability, and effective full-cell operation, making it a promising candidate for high-performance sodium-ion batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Entropy-Mediated Nonporous Polymer Separator Unlocks Ultra-Wide Temperature Range (−70°C to 150°C) Lithium Metal Batteries** — score: 0.800 The article discusses the development of a high-entropy polymer separator (HES) that enhances the performance and temperature adaptability of lithium metal batteries, allowing them to operate reliably across an extreme temperature range of -70°C to 150°C. The HES improves electrolyte wettability, thermal stability, and promotes efficient ion transport, while also forming stable interfaces that prevent dendrite formation and enhance battery longevity. The findings represent significant progress in separator technology and broaden the operational capabilities of lithium metal batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Stable and Low-Pressure Anode-Free All-Solid-State Li–S Batteries Enabled by Reaction-Derived Nanocrystalline-Amorphous Li2S Cathodes and Deformable Sodium Current Collectors** — score: 0.800 The article presents a low-pressure, anode-free all-solid-state Li-S battery design featuring a reaction-derived nanocrystalline-amorphous Li2S cathode and a flexible sodium current collector. This configuration enhances lithium ion transport and stability at solid-solid interfaces, resulting in exceptional cycling performance with nearly 100% capacity retention over 200 cycles at 4 MPa pressure, and significant stability even at pressures as low as 1 MPa. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Iron-mediated reversible lattice-oxygen redox enables stable 200 Wh kg−1 sodium-ion batteries** — score: 0.800 The authors present a novel approach using iron as a mediator to enhance the reversibility of lattice-oxygen redox processes in layered oxide cathodes, addressing the common issue of structural degradation in sodium-ion batteries. This method enables the achievement of stable energy densities of 200 Wh kg−1, significantly improving battery longevity and performance. Journal: *Nature Energy*
- **Ti3C2Tx/Ni5P4/NiP2 Double-Heterojunction Anodes With Fluoride-Free Interfacial Engineering for High-Rate and Ultrastable Lithium Storage** — score: 0.800 The article presents

a novel fluoride-free method for creating Ti3C2Tx/Ni5P4/NiP2 double-heterojunction anodes by using Lewis-acid molten salt etching and in-situ phosphorization. This composite exhibits exceptional lithium storage capabilities, achieving a capacity of 1172.4 mAh g−1 after 155 cycles and maintaining 487.1 mAh g−1 after 600 cycles, due to enhanced charge transfer and lithium ion diffusion facilitated by strong interfacial bonds. Journal: *Wiley: Small Methods: Table of Contents*

- **Ion-Adaptive Molecular Structure Facilitates Mg-Ion Dynamic for High-Capacity Aqueous Magnesium-Ion Batteries** — score: 0.800 This article presents an innovative ion-adaptive strategy utilizing flexible organic molecules (FOMs) to enhance the dynamic interaction of magnesium ions in aqueous magnesium-ion batteries (AMIBs). By modulating molecular steric hindrance, the study reveals how the innovative electrode structure can mitigate structural degradation during ion intercalation, leading to a record capacity of 332.1 mAh g−1 and impressive stability over 7,000 cycles, thus offering new avenues for improving electrode materials in multivalent-ion batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **From Top to Bottom: Manufacturing Process-Context Aware Resolution of Energy Device Electrodes Through a 3D Diffusion Generative Model** — score: 0.800 This article introduces a generative diffusion model that enhances the synthesis of 3D microstructures for electrodes in lithium-ion batteries and fuel cells. By incorporating a depth-aware augmentation pipeline, the model improves the accuracy of through-plane tortuosity factors, successfully replicating the expected performance of real electrodes and demonstrating potential for advancing the design of energy devices. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Hybrid Ionomer-Free Porous Transport Electrodes With Catalyst Coated Membranes for Enhanced Water Electrolysis** — score: 0.800 The study explores how coupling an ionomer-free porous transport electrode (PTE) with a traditional catalyst coated membrane (CCM) enhances water electrolysis performance in PEM systems. Using operando neutron radiography, it reveals that this hybrid configuration improves water distribution and catalyst utilization, leading to optimal interface contact and significantly lower ohmic overpotential during operation. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **When pathways trump orbitals in electrolyte design** — score: 0.800 The article discusses a new strategy in electrolyte design for high-voltage batteries that focuses on pathway selectivity to reduce electrolyte oxidation. By substituting vulnerable alpha-hydrogens rather than relying on traditional frontier orbital-based methods, the approach allows for stable performance at elevated voltages without the need for fluorination. Journal: *Nature Chemistry*
- **On-demand capacity extraction through thermo-electrochemical hysteresis in metastable Li-Cu alloy** — score: 0.700 The article discusses a method to enhance battery performance using a metastable lithium-copper alloy that stores lithium in an inactive state within the current collector. By triggering the release of this dormant lithium on demand, the approach mitigates capacity loss during cycling, ultimately improving both the capacity and cycle life of lithium-metal and lithium-ion batteries. This demonstrates the potential of metastable materials to add new functionalities to energy storage systems. Journal: *Joule*
- **A Fluorine-Free, Low Ion Pair Electrolyte for Transition Metal-Free and Halogen-Free Sodium Batteries** — score: 0.700 This study introduces a novel fluorine-free electrolyte chemistry for sodium-ion batteries that maintains high cycling stability and fast charging capabilities. By utilizing a sodium tetraphenylborate-diglyme electrolyte, the researchers achieved impressive performance metrics, including over 2000 cycles with minimal capacity loss and exceptional stripping-plating efficiencies with sodium metal anodes, paving the way for sustainable battery designs devoid of fluorinated components. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Molecular charge-topology electrolytes enable >630 Wh kg−1 lithium-metal batteries** — score: 0.700 The study introduces a molecular charge-topology framework for designing stable electrolytes that enhance the performance of lithium-metal batteries. This innovative approach minimizes undesirable reactions, facilitates efficient lithium cycling, and achieves impressive energy densities ex-

ceeding 630 Wh kg−1 in pouch cells, even under challenging conditions. Journal: *Joule*

- **Graphite edge architecture as a structural template for interphase evolution under fast charging** — score: 0.700 The article discusses the development of a graphite edge architecture that serves as a structural template, aiming to enhance the stability and performance of battery interphases during fast charging processes. The research highlights the potential of this innovative structure to improve energy storage efficiency and longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Electronic conductivity of solid electrolytes causes physical self-discharge in all-solid-state batteries** — score: 0.600 The study by Wang et al. reveals that the inherent electronic conductivity of solid-state electrolytes significantly contributes to physical self-discharge in all-solid-state batteries (ASSBs), posing a challenge to their longevity and performance. This finding highlights an important factor affecting the efficiency and lifespan of ASSBs. Journal: *Nature Energy*
- **Designing stratified solvation structure to address interfacial stability for high-voltage lithium metal batteries** — score: 0.600 The article discusses a novel approach to enhancing the interfacial stability of high-voltage lithium metal batteries by designing a stratified solvation structure. This innovation aims to improve battery performance and longevity by addressing challenges associated with lithium metal interfaces. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Unlocking Anion-Dependent Failure Mechanisms of Zinc Electrodes and Enhancing Cycling Stability via Asymmetric Interlayer** — score: 0.600 The article explores the anion-dependent failure mechanisms of zinc electrodes in energy storage systems, highlighting how these issues affect cycling stability. The authors propose the use of an asymmetric interlayer as a solution to enhance the longevity and performance of zinc-based batteries, detailing experimental findings that support their approach. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Regulating Lithium Ion Solvation via Dipole–Dipole Interactions for Fast Interfacial Kinetics in Ni-rich Layered Cathodes** — score: 0.600 The article discusses a novel approach to enhance the interfacial kinetics of Ni-rich layered cathodes in lithium-ion batteries by regulating lithium ion solvation through dipole–dipole interactions. This method aims to improve battery performance by facilitating faster ion movement at the cathode interface, potentially leading to more efficient energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **In Situ Electrochemically Driven A-Site Substitution Suppresses the Jahn–Teller Distortion for Long-Life Manganese-Based Prussian Blue Cathodes** — score: 0.500 The article discusses a novel method for enhancing the longevity of manganese-based Prussian blue cathodes by using in situ electrochemical techniques to perform A-site substitution, which effectively mitigates the Jahn-Teller distortion. This advancement promises improved performance for energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Modulating molecular and electronic structures of organic material through an electron-rich heterocyclic polymer strategy for high-performance anodic lithium storage** — score: 0.400 The article discusses a novel approach to enhancing the performance of organic materials for anodic lithium storage by utilizing an electron-rich heterocyclic polymer strategy. This method effectively modulates the molecular and electronic structures of the materials, promising significant improvements in energy storage capabilities. Journal: *ScienceDirect Publication: Nano Energy*
- **Anti-freezing cyclodextrin-modified cellulose eutectic hydrogel electrolytes for ultralong cycling low-temperature zinc-ion batteries** — score: 0.400 The article discusses the development of innovative anti-freezing cyclodextrin-modified cellulose eutectic hydrogel electrolytes designed for enhancing the performance of zinc-ion batteries. These electrolytes enable ultralong cycling capabilities even at low temperatures, addressing challenges in battery efficiency and longevity. Journal: *ScienceDirect Publication: Nano Energy*
- **Proton-specific recognition effect through confined-SCOF-functionalized highly selective hierarchical porous membrane for vanadium redox flow batteries** — score: 0.400 This arti-

cle discusses the development of a highly selective hierarchical porous membrane designed for vanadium redox flow batteries, which utilizes confined self-assembled coordination frameworks (SCOFs) for proton-specific recognition. The innovative membrane aims to enhance battery efficiency by selectively facilitating proton transport while blocking larger ions, thereby improving overall battery performance. Journal: *ScienceDirect Publication: Nano Energy*

- **Pushing Energy Density Limits: Anode-Less Solid-State Battery Integrated With Amorphous Lithium Vanadium Oxide Thin-Film Cathode (Adv. Energy Mater. 31/2026)** — score: 0.400 The article discusses the development of a high-performance anode-less solid-state thin film battery that utilizes an amorphous lithium vanadium oxide cathode and LiPON electrolyte. The authors highlight the importance of precise interface engineering with a Sn/C seeded current collector, which enhances lithium plating stability and contributes to achieving durable and high energy density in the battery design. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Understanding and suppressing gas evolution in lithium metal batteries with ether-based electrolytes** — score: 0.400 The article discusses a significant challenge in lithium metal batteries, namely gas evolution, particularly methane (CH4) resulting from the decomposition of ether-based electrolytes at the lithium anode. The authors propose a simple activation strategy that effectively mitigates these undesirable reactions, leading to reduced gas evolution and improved battery cycle life. Journal: *Nature Chemistry*
- **Rudolph A. Marcus obituary: theoretical chemist who explained how electrons jump between molecules** — score: 0.300 Rudolph A. Marcus, a renowned theoretical chemist, made significant contributions to understanding electron transfer between molecules, which underpins the functionality of batteries and solar cells. His pioneering work, which garnered him the Nobel Prize in Chemistry, has had lasting implications in the fields of energy storage and conversion. Journal: *Nature*
- **Reconstructing layered vanadium oxides with accelerated desolvation and electrostatic cation traps enables rapid Zn2+ storage at ultralow temperatures** — score: 0.300 This article discusses a method for reconstructing layered vanadium oxides to enhance Zn^2 storage capabilities at ultralow temperatures. The approach utilizes accelerated desolvation processes and electrostatic cation traps, leading to significant improvements in the efficiency and speed of energy storage in these materials. Journal: *ScienceDirect Publication: Nano Energy*
- **Lattice Engineering of Li2NiO2 via Metal Doping for High-Rate Prelithiation with Suppressed Gas Evolution** — score: 0.300 The article discusses advancements in Li2NiO2 through lattice engineering enhanced by metal doping, aimed at improving the material's performance in high-rate prelithiation applications. This approach not only boosts the efficiency of lithium-ion batteries but also minimizes harmful gas evolution during the process. Journal: *ScienceDirect Publication: Nano Energy*
- **Decoupling the Roles of B-Site Cation Defects and Oxygen Vacancies in Exsolved Perovskites for Robust CO2 Electrolysis in Solid-State Cells** — score: 0.300 The study investigates the specific roles of B-site cation defects and oxygen vacancies in exsolved perovskite materials, demonstrating that while both enhance electrical conductivity through improved covalency, excessive defects can reverse these benefits. By using in situ topotactic exsolution to refill B-site defects while maintaining oxygen vacancies, the authors achieve enhanced oxygen-ion transport and superior CO2 electrolysis performance, achieving high current densities without the formation of harmful SrCO3. This research highlights the importance of manipulating these defects to optimize perovskite electrocatalysts for solid-state CO2 electrolysis. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Enhancing Polysulfide Conversion via Heterogeneous Interfaces with Intensified Electron Transfer for High-Performance Lithium-Sulfur Batteries** — score: 0.300 The article discusses advancements in lithium-sulfur batteries, specifically focusing on enhancing the conversion of polysulfides through the use of heterogeneous interfaces that promote efficient electron transfer. This approach aims to improve the overall performance and viability of lithium-sulfur battery technology. Journal: *ScienceDirect Publication: Energy Storage Materials*